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When the Algorithm Meets the Absent Facility: The AI–Physical Execution Gap in India’s E-Waste Circular Economy

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Abstract

Artificial intelligence can now identify an end-of-life electronic device, grade its condition, estimate its residual value and rank its circular recovery pathways in under a second and at negligible marginal cost. In India, the world’s third-largest source of electronic waste, generating about 3.8 million metric tonnes a year under international accounting conventions and projected to approach 14 million tonnes by 2030, this inferential fluency has far outpaced the material substrate it assumes. The formal system holds roughly 322 registered recyclers and 72 registered refurbishers for a population of 1.4 billion; state-level processing capacity is extremely uneven; about 30 percent of volume by formal accounting, and considerably more by independent estimate, still moves through informal channels that have no digital interface; and rural India remains structurally disconnected from formal collection and recovery infrastructure. The argument of this chapter is that AI recommendation is not circularity. Using India’s semiconductor-dense device stream, and smartphones in particular, as its empirical anchor, the chapter develops the concept of the AI–Physical Execution Gap: the systematic divergence between the circular pathway an algorithm identifies as optimal and the subset of pathways a territory’s physical, institutional, logistical and market infrastructure can actually deliver. It decomposes the gap into five conditions (availability, accessibility, capacity, viability and legibility), proposes an Execution Feasibility Index and a Circularity Execution Ratio as measurement instruments, locates the binding constraint at each stage of the smartphone circular chain, formalises the reverse-logistics economics of the rural–urban divide, and asks whether India’s Extended Producer Responsibility and digital public infrastructure architectures narrow the gap or widen it. An illustrative computation under stated parameter assumptions indicates that consolidation, not haulage distance, dominates the viability of small-device reverse logistics by more than two orders of magnitude, and that consolidation alone does not restore feasibility where no licensed executor exists. The chapter closes by proposing Infrastructure-Constrained Circular AI, a design stance in which facility availability, distance, transport economics and real-time capacity enter the objective function as first-class constraints rather than as post hoc implementation friction.

Keywords: circular economy; artificial intelligence; electronic waste; India; reverse logistics; Extended Producer Responsibility; infrastructure constraints; emerging markets; semiconductor devices; rural–urban divide

1. Introduction: The Recommendation That Cannot Be Executed

Consider a five-year-old smartphone in a village in Nuapada district, Odisha. The screen is whole. The battery has degraded to about 60 percent of rated capacity. Its system-on-chip, camera module and radio-frequency front end all work. A well-trained recommendation model, fed with image data, diagnostic telemetry, secondary-market price signals and a materials database, will confidently return an optimal circular pathway: harvest the display assembly and camera module for the spare-parts market, replace the battery, refurbish the handset, and resell it into a price-sensitive segment where its residual utility is highest. The model will generate an expected recovery figure, an avoided-emissions figure and a confidence score. It will do so in under a second, at negligible marginal cost, and with defensible accuracy.

There is no registered refurbisher within several hundred kilometres of that village. There is no formal collection point within the block. The nearest authorised dismantler sits in a metropolitan cluster, where the logistics cost of moving a single 180-gram device is an order of magnitude higher than the value recoverable from it. The aggregator who might bundle the device into an economically viable shipment is informal, has no digital footprint, and is entirely invisible to the recommendation system. The device will almost certainly be sold for cash to a travelling scrap buyer, passed along an untraceable chain into an informal cluster, hand-dismantled for its high-value components, and its residual board fraction processed by methods that recover only part of the available metal content while externalising substantial environmental and occupational harm.

The recommendation was not wrong. It was simply irrelevant, and this chapter takes that irrelevance seriously as an object of study.

The literature on artificial intelligence and the circular economy has expanded rapidly and, for the most part, optimistically. Machine-vision sorting, predictive maintenance, digital product passports, residual-value estimation, demand forecasting for secondary materials and platform-mediated reverse marketplaces are all documented (Ellen MacArthur Foundation, 2019; Ghoreishi & Happonen, 2020; Sarc et al., 2019; Wilts et al., 2016). The theory of change underlying this work is usually implicit: better information will produce better circular outcomes, because if we can identify, classify, predict and optimise more accurately, materials will flow more efficiently through higher-value loops. That theory holds where the physical system is dense, capitalised and standardised enough to perform most of what an

optimiser might suggest. In India, the physical system does not satisfy that precondition, and the theory of change fails with it.

The claim advanced here is that inferential capacity has ceased to be the limiting factor in emerging-market circularity, that executional capacity now is, and that the two have decoupled. We call this decoupling the AI–Physical Execution Gap (hereafter AIPEG): the systematic and measurable difference between the set of circular pathways an algorithm identifies as optimal and the subset of those pathways that a given territory’s physical, institutional, logistical and market infrastructure can execute at acceptable cost and environmental impact. This is not an implementation detail to be absorbed by better change management. It is a structural property of the system, it varies predictably across geography and device class, and it is invisible to the very systems whose recommendations it nullifies.

The stakes of that invisibility are not academic. A programme can report improving circularity on the basis of recommendations issued, EPR certificates traded or platform-reported diversion while material outcomes stagnate or deteriorate. We call this phantom circularity: the accumulation of algorithmically legitimated circular intent that is never physically realised, yet is counted, reported and, under certificate-based compliance regimes, monetised.

India is the right place to make this argument, for four reasons. First, scale: it is the third-largest generator of e-waste in the world, and generation is growing faster than the infrastructure meant to absorb it. Second, asymmetry: its formal recovery capacity is concentrated in a small number of industrial clusters while its device population is national and majority rural. Third, institutional ambition: the E-Waste (Management) Rules, 2022, the CPCB EPR portal and India’s digital public infrastructure stack together constitute one of the most digitally mediated waste governance regimes in any emerging economy, which makes India a genuine test of whether digital coordination can substitute for physical capacity. Fourth, device composition: India’s e-waste stream is increasingly dominated by small, semiconductor-dense, high-value-per-kilogram devices, principally smartphones, whose circular economics are acutely sensitive to logistics cost, the variable current recommendation systems are least likely to represent.

1.1 Research questions

The chapter is organised around three questions.

- RQ1. How should the divergence between the circular pathway an unconstrained optimiser recommends for an Indian end-of-life smartphone and the pathway its surrounding infrastructure can actually execute be defined and measured?
- RQ2. Which of the executability conditions binds first, and does the binding condition change systematically with distance from a formal facility cluster?

- RQ3. Do India's certificate-based Extended Producer Responsibility regime and its digital public infrastructure stack narrow the execution gap, or does the design of either widen it?

Sections 2 and 3 answer RQ1 conceptually and formally. Sections 4 to 6 address RQ2 through the Indian material substrate, a stage-by-stage bottleneck analysis and a break-even computation. Sections 7 and 8 address RQ3 and derive the design consequences.

2. Conceptual Foundations: From Circular Ambition to Circular Capability

2.1 The R-Hierarchy and Its Silent Assumption

The circular economy is usually operationalised through a hierarchy of retention strategies (refuse, rethink, reduce, reuse, repair, refurbish, remanufacture, repurpose, recycle, recover) ordered by how much product function and embedded value each preserves (Kirchherr et al., 2017; Potting et al., 2017; Reike et al., 2018). The hierarchy is normative: higher R-strategies are preferred because they conserve more of the energy, labour and material complexity already embodied in the artefact. Geissdoerfer et al. (2017) and Blomsma and Brennan (2017) trace how a diffuse set of ideas consolidated into a governing paradigm, and Ghisellini et al. (2016) survey its diffusion across policy regimes.

The hierarchy does not tell us who performs each R, where, at what cost, or at what throughput. It ranks desirability and takes availability for granted. In dense industrial economies that assumption is close enough to the truth that it rarely surfaces as an analytical problem. In India it is wrong in a specific and consequential way: across most of the national territory the higher R-strategies have no local executor. In practice the ten-item hierarchy collapses into a two-item menu of informal recycling or landfill, whatever the artefact merits and whatever an algorithm recommends.

Velenturf and Purnell (2021) observe that circular-economy scholarship has systematically prioritised material flows over the institutional and infrastructural systems that carry them, and Corvellec et al. (2022) argue that the circular economy functions partly as a discursive device that displaces attention from the material constraints on implementation. We propose a constructive extension of both positions. Circularity is not illusory; the field simply lacks an explicit variable for executability, and in emerging markets that omission is not a rounding error but the dominant term.

2.2 What AI Actually Does in the Circular Economy

Applications of AI to circular systems fall into four functional families, each standing in a different relation to physical infrastructure.

- Identification and classification. Convolutional and transformer-based vision models classify device type, brand, model and cosmetic condition from images, while spectroscopic and X-ray systems classify material fractions on sorting lines (Gundupalli et al., 2017; Nowakowski & Pamuła, 2020; Sarc et al., 2019). These systems are substrate-coupled: they operate inside a facility and are useful because the facility exists.
- Prediction and forecasting. Models estimate remaining useful life, residual market value, component failure probability and secondary-material demand. Prediction is substrate-independent. It can be performed anywhere, on any device, at negligible cost, and its accuracy does not depend on whether a facility is present.
- Routing and optimisation. Systems assign end-of-life devices to R-pathways, plan collection routes, allocate volumes to facilities and price transactions. This family is nominally substrate-aware, but in the dominant literature and in commercial practice it treats the facility network as an exogenous given: usually complete, always reachable.
- Coordination. Platforms, digital product passports, marketplaces and traceability systems match the supply and demand of participating actors (Jensen et al., 2023). Coordination is substrate-conditional: it raises the productivity of existing capacity but cannot create capacity that does not exist.

The asymmetry across these families is the crux of the argument. The substrate-independent capabilities, which are the ones that scale fastest and most cheaply, have advanced dramatically. The substrate-coupled and substrate-conditional capabilities are hard-bounded by physical assets that scale slowly, require capital and are unevenly distributed in space. An AI system could classify every smartphone retired in India this year. India's formal system can process a small fraction of that volume, and only in a few places. The recommendation layer has scaled by orders of magnitude. The execution layer has not.

2.3 The Missing Variable: Executability

We therefore introduce the construct of executability explicitly. A circular recommendation is a triple: a device state, a proposed retention strategy and a designated executing facility. A recommendation is executable if and only if five conditions hold simultaneously.

- Availability — a facility exists within the reachable system that is technically equipped and licensed to perform the strategy.
- Accessibility — the device can physically reach that facility along an existing, lawful and functioning logistics chain.

- **Capacity** — the facility has throughput available to accept the device within a window before its residual value decays below threshold.
- **Viability** — recovered value net of collection, aggregation, transport, processing and compliance cost is positive for every actor whose participation the chain requires.
- **Legibility** — the transaction is recordable, verifiable and creditable within the governing institutional framework, so that the actors who perform it are recognised and compensated.

If any condition fails, the recommendation is invalid, however accurate the underlying inference. Executability is therefore a conjunctive rather than a compensatory construct: excellence on four dimensions cannot offset failure on the fifth. That conjunctive structure is what makes the AI–Physical Execution Gap severe rather than merely inconvenient, and it is the formal reason why marginal improvements in model accuracy yield near-zero marginal improvement in circular outcomes once any one condition binds.

3. Defining the AI–Physical Execution Gap

3.1 Conceptual Definition

We define the AI–Physical Execution Gap as the systematic divergence between the set of circular pathways an algorithmic system identifies as optimal for a population of end-of-life devices and the subset of those pathways that the surrounding physical, institutional, logistical and market infrastructure can execute at acceptable cost and environmental impact within the devices' value-retention window.

The definition has four analytically important properties. The gap is systematic rather than stochastic: it is structured by geography, device class and institutional design, and is therefore predictable and addressable rather than random. It is relational: it is a property of the pairing of an algorithm with a territory, not of either alone, so that the same model exhibits a small gap in a metropolitan cluster and a near-total gap two hundred kilometres away. It is time-bounded: value decays, so a pathway executable only after the value-retention window has closed was never executable in any meaningful sense. And it is invisible by default: nothing in a standard recommendation architecture emits a signal when a recommendation goes unacted upon, so the gap can widen without limit while reported performance improves.

3.2 Five Dimensions of the Gap

The five conditions for executability map onto five dimensions of the gap, each with characteristic causes, observable indicators and remediation instruments.

Table 1. *The five dimensions of the AI–Physical Execution Gap.*

Dimension	Failure condition	Primary observable indicator	Dominant cause in India	Remediation instrument
Availability	No licensed facility can perform the strategy anywhere in the reachable system	Facility counts per device-population unit; strategy coverage ratio	Extreme scarcity of registered refurbishers relative to recyclers; near-absence of component-level recovery licensing	Licensing reform; capacity investment; strategy-specific authorisation categories
Accessibility	A facility exists but the device cannot lawfully or economically reach it	Mean and 90th-percentile distance to nearest capable facility; collection-point density	Spatial concentration of capacity in industrial clusters; absent rural collection infrastructure	Aggregation nodes; hub-and-spoke reverse logistics; postal and retail network integration
Capacity	The facility exists and is reachable but cannot accept the volume in time	Facility utilisation rate; queue time versus value-decay rate	Thin installed throughput; seasonal and campaign-driven volume spikes; no real-time capacity signalling	Capacity registers; dynamic allocation; buffer storage nodes
Viability	Execution is physically possible but value-negative for at least one required actor	Net margin per device by pathway and origin; break-even distance and minimum viable consignment	High fixed consignment cost for small devices; low formal buy-back prices versus informal cash offers	Floor prices; EPR fee modulation; origin-indexed subsidy; consolidation economics
Legibility	Execution occurs but cannot be recorded, verified or credited	Share of flow with digital provenance; informal-sector integration rate	Structural exclusion of informal aggregators from the EPR portal architecture	Tiered registration; aggregator digital identity; verified handover protocols

The dimensions are ordered by the position in the chain at which they bind, and they combine multiplicatively rather than additively. Better logistics does not repair an availability failure. Additional facility capacity does not resolve an accessibility failure at a location the device cannot reach. Information does not remedy a viability failure. The measurement approach developed below captures this multiplicative structure directly, and it explains a persistent puzzle for policymakers: large investments along any single dimension often produce negligible aggregate improvement, because some other dimension remains binding.

3.3 Operationalisation: The Execution Feasibility Index

Let d be an end-of-life device with state vector $s(d)$. Let r be a candidate retention strategy in the R -hierarchy, and let $F(r)$ be the set of licensed and equipped facilities able to perform r . The Execution Feasibility Index $\varphi(d, r)$ is defined as the product of five terms, each bounded on the unit interval:

$$\varphi(d, r) = A(r) \times Acc(d, r) \times Cap(r, t) \times V(d, r) \times L(d, r) \quad (1)$$

where the terms are as follows.

- $A(r)$ is the availability term, a binary indicator equal to 1 if $F(r)$ is non-empty within the administratively reachable system and 0 otherwise. The binary form is deliberate. Note that the

existence of a licensing category is not the same as the availability of a facility able to accept a particular device, so $A(r)$ as specified here is a permissive upper bound.

- $Acc(d, r)$ is the accessibility term, a monotonically decreasing function of the effective transport distance from the device origin to the nearest facility in $F(r)$, modulated by route quality, consolidation opportunity and lawful transport constraints on hazardous fractions. We adopt a logistic-decay form, $Acc = 1 / (1 + \exp[\beta(\delta - \delta_{1/2})])$, where δ is effective road distance, $\delta_{1/2}$ is the distance at which accessibility falls to one half, and β controls the steepness of decay. Both parameters are device-class specific: decay is steep for low-mass, low-unit-value items and shallow for high-value or high-mass items over which transport cost is amortised against recoverable value.
- $Cap(r, t)$ is the capacity term, the probability that a facility in $F(r)$ can accept the device within the value-retention window beginning at time t . It is computed from installed throughput, current utilisation and expected queue length. It is the term most amenable to real-time instrumentation and the one most conspicuously absent from current systems.
- $V(d, r)$ is the viability term, a normalised measure of the net economic surplus generated by executing r on d after collection, aggregation, transport, processing, compliance and margin requirements at every intermediate node. V must be read as the minimum across the chain, not the sum across it: a chain with positive aggregate surplus but one value-negative intermediary will not run, because that intermediary defects to the informal channel. Formally, V is zero if any required actor faces a non-positive surplus.
- $L(d, r)$ is the legibility term, the probability that a completed transaction will be recorded and credited within the prevailing institutional framework. It is zero along routes passing through unregistered actors. This is why, in India, legibility failure and viability success are frequently correlated: the least legible path is often the cheapest executable one.

3.4 Aggregate Measurement: The Circularity Execution Ratio

The Execution Feasibility Index evaluates a single device–strategy pair. System-level assessment requires an aggregate. Let $R^*(d)$ be the strategy an unconstrained optimiser would recommend for device d on purely technical grounds, that is, the highest attainable point on the R -hierarchy. Let $\hat{R}(d)$ be the highest-ranked strategy whose ϕ exceeds a decision threshold. For a population of devices D we define the Circularity Execution Ratio as:

$$CER(D) = [\sum_{\{d \in D\}} w(d) \cdot value(\hat{R}(d))] / [\sum_{\{d \in D\}} w(d) \cdot value(R^*(d))] \quad (2)$$

where $w(d)$ weights devices by mass, embodied impact or recoverable value according to the purpose of the analysis, and $value(\cdot)$ maps a strategy to a retained-value or avoided-impact score. A system with no

execution gap has a CER of one; a system whose recommendations are systematically inexecutable has a CER approaching zero. A summary measure of the AI–Physical Execution Gap follows directly as $AIPEG(D) = 1 - CER(D)$.

Three properties make this measure useful for policy. It is decomposable: because ϕ is multiplicative, shortfalls can be attributed to specific dimensions, allowing a state or district to identify which constraint to relieve first. It is spatially disaggregable: computing CER by district reveals the geography of inexecutable circularity, and mapping that geography identifies where marginal infrastructure investment would release the greatest volume, arguably the single highest-value application of AI in an infrastructure-constrained circular economy. And it is counterfactual: it measures realised outcomes against technical potential rather than against last year's performance, which is the relevant comparison for a resource-constrained transition.

3.5 The Status of These Measures

ϕ and CER are proposed measurement instruments, not validated indices. Their parameters, particularly the accessibility decay parameters β and $\delta^{1/2}$, the consolidation factor and the normalisation of V , require calibration against operator-level logistics and facility data that are not currently in the public domain. Section 6 computes both quantities for a small set of stated parameter assumptions in order to demonstrate the structure of the framework and the relative magnitude of its terms. Those computations are illustrative. They are not empirical estimates of Indian circularity, and they should not be cited as such. Section 9.4 sets out what a first empirical estimation would require.

4. India's Material Substrate: What the Algorithm Is Recommending Into

4.1 Volume, Composition and Trajectory

India's e-waste generation is approximately 3.8 million metric tonnes per annum under international monitoring conventions, making it the third-largest generator in the world after China and the United States (Baldé et al., 2024). Generation is projected to reach roughly 14 million tonnes by 2030 on a business-as-usual path of device penetration, income growth, shortening replacement cycles and the electrification of formerly non-electronic product categories.

For the argument developed here, the composition of that stream matters more than its size. India has recorded among the fastest growth rates in the world in screens, computing equipment and small information and telecommunications equipment: the fraction that is semiconductor-dense, low in unit mass, high in unit value and rich in critical and precious metals. The country has more than a billion active mobile connections, and aggressive device financing and trade-in promotion are shortening handset

replacement cycles. A smartphone weighs roughly 180 to 200 grams and contains gold, silver, palladium, copper, cobalt, lithium, tantalum and rare-earth elements at concentrations that make its printed circuit board assembly one of the most metal-dense materials in the anthropogenic environment, richer per tonne than primary ore by orders of magnitude. It is precisely the device class that most rewards high-R strategies and most penalises long-distance, low-density reverse logistics. The economics of smartphone circularity are consequently governed by the variable that recommendation systems are least likely to represent.

4.2 Evidence of Data Asymmetry

A striking feature of the Indian e-waste evidence base is that its credible sources diverge by a factor of three. International monitoring, which estimates generation from apparent consumption and product lifespan distributions, puts annual arisings in India close to 4 million tonnes. India's own official figures, derived from registered producers' sales data and prescribed average equipment lifespans and published through the CPCB EPR portal, recorded approximately 1.25 million tonnes in FY 2023–24 and 1.40 million tonnes in FY 2024–25, with official collection-and-processing shares of roughly 62 percent and 71 percent respectively against those figures. Independent assessments have repeatedly found that the unorganised sector handles the bulk of Indian e-waste, commonly estimated at between 70 and 90 percent.

These are not competing errors to be adjudicated. The divergence is itself the most direct evidence for the argument of this chapter. The formal system measures against the official denominator; the country loses material against the international one. The ratio between them is a crude but usable proxy for the legibility dimension of the execution gap. That reported formal processing rates climb steeply against the smaller denominator while independent estimates of informal processing remain flat is a textbook case of phantom circularity: improvement in the measurement apparatus read as improvement in material outcomes. Any AI system trained on portal data inherits this bias structurally. It learns what the formal system can see, optimises within that boundary, and is blind by construction to most of the material it purports to govern.

4.3 Composition and Scarcity of Facilities

The formal execution layer for this volume comprises roughly 322 registered recyclers and 72 registered refurbishers. Registration counts on the CPCB EPR portal are dynamic and have ranged between approximately 300 and 450 authorised dismantlers and recyclers across recent reporting cycles, depending on category definition and reporting date. The composition, however, is stable, and it is the composition that carries the analytical interest.

Three features of that composition matter. First, absolute density is extraordinarily thin: a few hundred facilities serve a device population numbering in the billions, a facility-to-population ratio at least an order of magnitude below that of jurisdictions with comparable circular ambition. Second, there are roughly 4.5 recyclers for every refurbisher, which means the formal system is structurally biased toward the lower end of the R-hierarchy. This is the availability dimension expressed as systematic distortion rather than random shortage: even were a perfect optimiser to recommend refurbishment at scale, the corresponding execution layer would be roughly four and a half times thinner than the layer for material recycling. Third, licensing categories are defined around dismantling and material recovery rather than component-level harvesting, repair-grade parts qualification or module-level remanufacturing. The intermediate R-strategies, which retain the most value for semiconductor devices, have no dedicated regulatory home and therefore no reliable execution layer.

Capacity is also spatially concentrated. Registered recycling facilities are present in only a minority of states, and installed processing capacity is heavily concentrated in a small number of industrial regions that handle a disproportionate share of national throughput. Generation, by contrast, is national in scope: a large share of arisings originates outside the metropolitan cores that host processing capacity, and a substantial share originates in districts several hundred kilometres from any licensed facility. This mismatch between where devices are retired and where they can be processed is the material basis of the accessibility dimension.

4.4 The Shadow Execution Layer: The Informal Sector

By formal accounting, about 30 percent of India's e-waste is processed through informal channels; by independent estimate, considerably more. Analytically, the decisive point is not the percentage but the functional role of the sector. The informal sector is not the absence of an execution layer. It is a second execution layer, with a materially better profile on three of the five executability dimensions and a catastrophic profile on the remaining two.

It performs well on accessibility: in dense urban settlements and in the small towns and villages where no formal collection network exists, itinerant buyers reach households at the doorstep. It performs well on viability, paying cash on the spot at prices formal buy-back schemes seldom match, because its cost structure excludes compliance, environmental control, formal labour cost and taxation. It performs well on capacity, because the network is elastic and absorbs volume spikes without capital investment. It performs close to zero on legibility: transactions leave no digital trace, and the actors are structurally excluded from a portal-based EPR architecture that presupposes registration, documentation and formal invoicing. And it fails on the environmental and occupational dimensions that belong to any serious definition of circular quality, since open burning, acid leaching and manual desoldering recover part of

the available metal content while imposing severe health externalities on workers and surrounding communities (Chatterjee & Kumar, 2009; Wath et al., 2011; Awasthi & Li, 2017).

This yields an uncomfortable but important strategic conclusion. Across much of India the informal sector is the only actor capable of performing collection and aggregation at all. An AI architecture that treats informality purely as a compliance problem to be eliminated will produce recommendations that cannot be executed anywhere outside a handful of metropolitan clusters. An architecture that treats it instead as an execution asset to be selectively formalised, preserving its accessibility and viability advantages while progressively substituting its processing functions, faces a materially different feasibility frontier. Sections 7 and 8 develop that design choice.

5. Stage-by-Stage Bottleneck Analysis: The Smartphone Circular Chain

5.1 Stage 1: Discard Decision and Collection

At this stage an AI system estimates residual value, prompts the household toward a formal channel and offers a buy-back price. Execution requires either a formal collection point within a tolerable travel distance or a reverse-pickup service quoting a price better than the informal alternative. Outside metropolitan areas neither condition is normally satisfied: formal collection points are sparse, take-back is concentrated in organised retail and brand service networks that are themselves urban, and formal quotes are systematically undercut by immediate cash offers. Hoarding compounds the problem, with a large share of retired handsets sitting in households for years and losing residual value while waiting for a channel. The binding dimensions are accessibility and viability, and the failure occurs upstream of any downstream capacity constraint.

5.2 Stage 2: Aggregation and Consolidation

For a 180-gram device, reverse logistics is economical only for consolidated loads. The AI routing engine optimises pickup routes and consolidation points. Execution requires physical aggregation nodes with storage, basic security and a custody protocol. Across most of the country, formal aggregation infrastructure below the district level is sparse or absent, and the actors who do perform aggregation, namely informal kabadiwalas and small traders, sit outside the digital system and therefore outside the optimiser's world model. This is the most consequential structural gap in the chain: it is where accessibility failure and legibility failure compound, and where most Indian e-waste irreversibly leaves the formal pathway.

5.3 Stage 3: Triage and Grading

Machine vision and diagnostic software permit initial R-strategy assignment, cosmetic condition grading and component functionality verification. This is the most substrate-coupled stage, where AI performs best and adds the most demonstrable value. It requires a facility with imaging apparatus, diagnostic rigs and trained operators. India's formal triage capacity is largely confined to a small number of refurbishment and trade-in operations. There is no national harmonisation of grading standards, so a grade assigned at one node cannot be interpreted as equivalent to a grade assigned at another node. This is a legibility failure at the technical rather than the institutional level, and a direct obstacle to distributed triage.

5.4 Stage 4: Reconditioning and Refurbishment

Refurbishment is almost always the value-maximising strategy for a functional handset with a degraded battery, because it retains the device's full embodied impact. It requires genuine or qualified spare parts, service documentation, diagnostic access and, for lawful resale under warranty, an authorised entity. Availability binds immediately and hard: there are 72 registered refurbishers nationally. Manufacturers' parts policies further restrict the supply of spares for older models, and repairability varies considerably across device designs. India's evolving right-to-repair framework bears directly on this stage, because it acts on precisely the constraint that determines whether the highest-value R-strategy is executable at all.

5.5 Stage 5: Component Harvesting

Where full refurbishment is not feasible, salvaging display assemblies, camera modules, connectors and memory for the repair market yields substantially more value than shredding. Component-level value estimation is the kind of pattern-recognition problem at which models excel, and it is among the strategies AI systems most readily recommend. It is also the strategy with the thinnest formal implementation in India: harvesting occupies a regulatory interstice between dismantling and refurbishment, has no dedicated authorisation category, and has no qualification standard that would allow harvested parts to enter formal repair channels under warranty. Informal clusters, meanwhile, harvest components at scale and with considerable skill. The capacity exists, but it is illegible and unqualified, and is therefore unavailable to the formal system.

5.6 Stage 6: Dismantling and Fraction Separation

Manual and semi-automated dismantling separates battery, display, board, housing and cable fractions for divergent downstream processing. This is where India's formal capacity is genuinely concentrated and where authorised dismantlers operate. The binding constraints here are capacity and throughput rather than availability, together with a technical constraint rarely encoded in recommendation systems:

lithium-ion batteries are classified as hazardous for transport, and devices containing them are subject to routing restrictions that a naïve distance-minimising optimiser will violate.

5.7 Stage 7: Refining and Material Recovery

Recovering gold, silver, palladium, copper and cobalt from board fractions requires hydrometallurgical or pyrometallurgical capability, a technical standard only a fraction of registered recyclers meet. India's integrated capacity to refine high-value precious metals from electronic scrap is limited, and part of the highest-value fraction is exported for refining, an outcome that satisfies material circularity in the global aggregate while forgoing domestic value capture. Rare-earth elements are recovered from Indian e-waste at a negligible rate, consistent with the global pattern in which recovery meets on the order of one percent of demand. At this stage the binding constraints are capacity and viability, and the limiting factor is the sophistication of installed technology rather than the number of registered facilities.

5.8 Stage 8: Resale of Recovered Materials and Refurbished Devices

The circular loop closes only when recovered materials and refurbished devices re-enter production and consumption. India has a large and growing secondary smartphone market, which is a genuine structural advantage: demand for value-retained devices is not the limiting factor. The constraints here are trust and standardisation. A refurbished handset without a credible grade, warranty and provenance record attracts a steep discount, and that discount cascades backward through the chain, destroying the viability of the refurbishment pathway that produced the device. A digital product passport architecture would generate a disproportionate return at this stage precisely because it addresses a legibility failure at the point where value is realised rather than where material is handled.

5.9 Synthesis

Table 2. Stage-wise comparison of algorithmic capability and formal execution capability in India's smartphone circular chain.

Stage	AI capability maturity	Formal execution capability	Binding gap dimension
1. Discard and collection	High	Low outside metropolitan areas	Accessibility, Viability
2. Aggregation	High	Very low; informally dominated	Accessibility, Legibility
3. Triage and grading	Very high	Low; unstandardised	Availability, Legibility
4. Repair and refurbishment	Moderate	Very low (72 registered entities)	Availability
5. Component harvesting	High	Near zero formally; high informally	Availability, Legibility
6. Dismantling	Moderate	Moderate; spatially concentrated	Capacity, Accessibility
7. Material recovery	Moderate	Limited at high-value tiers	Capacity, Viability
8. Resale	High	Moderate and improving	Legibility

Read across the table and a consistent, diagnostically important pattern emerges: algorithmic capability is highest exactly where formal execution capability is lowest. Triage, harvesting and residual-value estimation, the tasks models perform best, depend on refurbishment and harvesting layers that barely exist in India's formal system. Conversely, the stages where the marginal AI contribution is smallest, dismantling and bulk material recovery, are the stages where India holds real capacity. The execution gap is not spread evenly along the chain; it is concentrated where the algorithmic promise is greatest. Any intervention strategy that overlooks this inverse relationship will misdirect both computational and capital investment.

6. The Urban–Rural Execution Divide

6.1 Distance, Mass and Consolidation

Whether circularity is worth performing on a low-mass, high-value device is governed by the cost of moving that device to a capable facility. That cost has two components with very different behaviour. One is a haulage component, roughly proportional to distance and to mass. The other is a fixed component per consignment (vehicle call-out, stop time, handling, custody and documentation) that is incurred once per shipment regardless of how many devices the shipment carries, and is therefore divided across the consolidation factor. Recoverable value per device, meanwhile, is bounded and modest for a mid-range end-of-life smartphone. There is consequently a break-even distance beyond which moving a device costs more than the value recoverable from it and, independently, a distance beyond which transport emissions exceed the emissions avoided by the circular intervention. Above either threshold the recommendation is not merely uneconomic but counterproductive.

This is the divide the present section analyses. Rural circularity is not simply a harder version of urban circularity; it is structurally different, because the same device generates a different cost structure depending on whether a consignment can be assembled around it. An optimiser that represents neither distance nor consolidation will systematically recommend value-destroying actions across most of India's geography while reporting them as circular successes.

6.2 A Break-Even Analysis

Let the net circular surplus of routing one device from an origin to a capable facility be:

$$S = V_{rec} - C_{acq} - C_{agg} - C_{proc} - C_{comp} - C_{cons}/n - c_t \cdot \delta \cdot m \quad (3)$$

where V_{rec} is recoverable value, C_{acq} the acquisition price paid to the holder, C_{agg} aggregation and handling cost, C_{proc} processing cost, C_{comp} compliance and documentation cost, C_{cons} the fixed

cost of assembling and dispatching one consignment, n the consolidation factor (devices per consignment), c_t the haulage cost per tonne-kilometre, δ the effective road distance and m the device mass. Writing $M = V_{\text{rec}} - C_{\text{acq}} - C_{\text{agg}} - C_{\text{proc}} - C_{\text{comp}}$ for the margin before transport, the break-even distance is:

$$\delta_{\text{be}} = (M - C_{\text{cons}}/n) / (c_t \cdot m) \quad (4)$$

and the minimum viable consignment size at a given distance is:

$$n_{\text{min}} = C_{\text{cons}} / (M - c_t \cdot \delta \cdot m) \quad (5)$$

Appendix A derives (4) and (5) from (3). Note that (3) departs from the formulation most commonly used in reverse-logistics costing, which expresses transport cost purely as a tonne-kilometre term. For a 180-gram device that formulation is dimensionally correct but empirically misleading, because the mass-distance product is so small that the haulage term becomes negligible against the per-consignment fixed cost. Separating the two components is what allows the model to identify the operative constraint rather than the nominal one.

Four structural properties follow for any reasonable parameterisation.

- First, the break-even distance increases with the consolidation factor but is bounded above by $M/(c_t \cdot m)$. For a smartphone that bound is very large. Distance therefore does not bind through haulage cost; consolidation binds, and distance matters chiefly because it determines whether a consignment can be assembled at all within the value-retention window.
- Second, at constant value the break-even distance is inversely proportional to device mass, which explains why smartphones are simultaneously the most attractive devices to recover and among the most fragile to route: their high value density supports long-distance movement when consolidated and collapses to a near-zero viable radius when they are handled individually.
- Third, consolidation cannot rescue a pathway whose margin before transport is non-positive. Where $M \leq 0$, expression (5) admits no finite n_{min} . Aggregation repairs viability failures caused by cost allocation; it does not repair viability failures intrinsic to the pathway's economics.
- Fourth, C_{comp} enters with the same sign as transport cost. Every additional layer of documentation, verification and compliance imposed on a formal pathway contracts the region in which that pathway is viable and expands the region in which the informal pathway is the only economically rational choice. Compliance design is, unintentionally, spatial policy.

6.3 Three Territorial Regimes: An Illustrative Computation

To show what these properties imply, we evaluate (3) and (1) for three territorial regimes and three retention strategies under the parameter assumptions set out in Appendix B. The parameters are plausible order-of-magnitude values drawn from published freight tariffs and secondary-market prices; they are not measured operator data, and the results below are a demonstration of structure, not an estimate of Indian circularity.

Table 3. Illustrative computation of per-device net surplus and Execution Feasibility Index by territorial regime and retention strategy, under the parameter assumptions of Appendix B. Values in Indian rupees per device. These are illustrative computations, not empirical estimates.

Territorial regime	δ (km)	n	Transport ₹/device	Strategy	M (₹)	S (₹)	Binding term	ϕ
Metropolitan core	25	200	2.03	Refurbishment	850	848	—	0.70
Metropolitan core	25	200	2.03	Component harvesting	200	198	Availability	0.00
Metropolitan core	25	200	2.03	Material recycling	-35	-37	Viability	0.00
Mid-periphery	180	40	22.71	Refurbishment	850	827	—	0.48
Mid-periphery	180	40	22.71	Component harvesting	200	177	Availability	0.00
Mid-periphery	180	40	22.71	Material recycling	-35	-58	Viability	0.00
Hinterland, no node	520	1	250.59	Refurbishment	850	599	Accessibility	0.01
Hinterland, no node	520	1	250.59	Component harvesting	200	-51	Availability	0.00
Hinterland, no node	520	1	250.59	Material recycling	-35	-286	Viability	0.00
Hinterland with node	520	60	23.93	Refurbishment	850	826	—	0.51
Hinterland with node	520	60	23.93	Component harvesting	200	176	Availability	0.00
Hinterland with node	520	60	23.93	Material recycling	-35	-59	Viability	0.00

Three findings follow. First, across the full range examined, the haulage term contributes between ₹0.03 and ₹0.59 per device, a spread of about ₹0.56 across 495 kilometres, while the consolidation term contributes between ₹2.00 and ₹250.00, a spread of ₹248. The consolidation factor dominates haulage distance by more than two orders of magnitude. The widely repeated claim that distance is the primary obstacle to rural circularity for small devices is, on this parameterisation, incorrect in its mechanism even though it is correct in its conclusion: what makes remote devices unroutable is that no consignment can be assembled around them, not that they are far away.

Second, inserting an aggregation node in the hinterland raises the per-device net surplus of the refurbishment pathway from ₹599 to ₹826 and, more importantly, raises its Execution Feasibility Index

from approximately 0.01 to approximately 0.51, because consolidation brings a distant but existing refurbisher into the effectively reachable system. This is the quantitative case for the aggregation node as the highest-leverage infrastructure asset in rural circularity.

Third, and against the grain of that conclusion, consolidation does nothing for component harvesting. Its net surplus is comfortably positive in three of the four regimes, yet its Execution Feasibility Index is zero everywhere, because the availability term is zero: no authorisation category exists for component harvesting anywhere in India. Similarly, formal material recycling of an individual handset is value-negative in every regime, and no consolidation factor repairs it; registered recyclers sustain themselves on mixed bulk streams and certificate revenue rather than on per-handset margin. Aggregation infrastructure and licensing reform are therefore complements, not substitutes. Building nodes without opening a harvesting category converts an accessibility failure into an availability failure and leaves ϕ at zero.

This is the answer to RQ2, and it is conditional and uncomfortable. Under present institutional arrangements an AI recommendation issued for a rural end-of-life device cannot be executed at a cost and environmental impact that make circularity worthwhile, unless an aggregation layer is interposed that converts individual devices into consolidated flows before the transport cost is allocated, and unless a licensing category exists for the strategy the optimiser selects. Either condition alone leaves the recommendation inexecutable.

6.4 Design Implications

This inverts the conventional intervention logic. The standard prescription for rural circularity is to push formal collection and processing capacity outward from the core. The break-even structure shows that to be the more expensive of two options: because per-device transport cost falls as C_{cons}/n , low-capital aggregation nodes, which require storage, a custody protocol, a digital identity and a payment rail but no processing plant, extend the viable radius far more cheaply than duplicating processing capacity. Such nodes can be hosted within existing rural physical networks of national coverage, including the postal network, cooperative structures, common service centres and organised retail.

It also redefines what AI should be doing in rural India. Per-device pathway recommendation is not a useful computation there, because it is precisely the computation whose output cannot be acted upon. The useful computation is aggregation-node siting and consolidation scheduling: where to place a node, at what cost, to convert the largest volume of currently inexecutable recommendations into executable volume, and when to dispatch once accumulated volume makes dispatch worthwhile. That is a

facility-location and inventory-routing problem under uncertainty, a class of problems with mature methods, and one tractable with data India already collects.

7. Institutional, Policy and Market Conditions for Closing the Gap

7.1 What the E-Waste (Management) Rules, 2022 Cover, and What They Do Not

India's current regime, notified in November 2022 and effective from April 2023, is genuinely ambitious. It substantially expands product coverage, establishes a centralised CPCB portal for registration and filing, sets rising collection targets across compliance years, and creates a tradable EPR certificate issued to registered recyclers on the basis of quantities recycled and purchased by producers to discharge their obligations. It is, in effect, a digital platform for circular compliance.

Mapped against the five dimensions, the regime's coverage is highly uneven. It takes legibility seriously and creates a national data spine for registered actors where none existed. It addresses viability partially and indirectly, since certificate revenue improves recycler economics. It does not address availability, because nothing in a certificate market requires facilities to be sited in underserved areas or in underserved licensing categories. It does not address accessibility, because certificates are fungible across space: a certificate generated in a high-capacity industrial state discharges an obligation arising from devices sold in a state with no facilities at all. And it does not address capacity signalling, because the portal records completed transactions rather than available throughput.

Two structural consequences follow, and both amplify the execution gap directly. First, because certificates are spatially fungible, there is no incentive to build capacity where the devices actually are. A producer discharges its obligation most cheaply by buying certificates from the lowest-cost recycler, which is located where capacity is already concentrated. The compliance market therefore actively rewards spatial concentration: the accessibility dimension is not merely ignored but structurally aggravated. Second, certificate-based compliance is measured in processed tonnage rather than retained value, and is therefore indifferent to whether an end-of-life device is refurbished or shredded. A regime that rewards mass recycling cannot preferentially incentivise the higher R-strategies, a plausible institutional explanation for the four-and-a-half-to-one ratio of recyclers to refurbishers documented in Section 4.3. The availability distortion is not incidental; it is the equilibrium outcome of the incentive design.

7.2 Digital Public Infrastructure as a Coordination Substrate

India's digital public infrastructure is the most credible existing asset for addressing the legibility and viability dimensions together: a population-scale identity layer, a real-time low-cost payment rail, a

consent-based data-sharing architecture and an open network protocol for digital commerce. Its relevance to circularity remains largely unexplored.

The mechanism is specific. The exclusion of informal aggregators from the EPR system is not a matter of intent but of transaction infrastructure: registration is costly, invoicing presumes formal accounting, and a handover cannot be paid for and evidenced at low cost. A tiered digital identity for aggregators, instant account-to-account settlement on handover, and a lightweight cryptographic handover receipt would convert an illegible transaction into a legible one without requiring the aggregator to re-engineer their business. An open network protocol layer could then publish facility capacity, current buy-back prices and accepted device categories as discoverable machine-readable offers: exactly the data the capacity and viability terms of the Execution Feasibility Index require, and exactly the data no current system publishes.

The caveat is the one this chapter exists to press. Digital public infrastructure improves coordination, and coordination raises the productivity of existing capacity. It does not create capacity. A perfectly instrumented market containing 72 registered refurbishers will allocate those 72 refurbishers efficiently and will still be unable to refurbish at national scale. Digital coordination is necessary; it is plainly not sufficient.

7.3 Five Institutional Conditions

In summary, the execution gap can be compressed only by the joint application of five institutional conditions.

- Spatially differentiated compliance. Certificate value should be modulated by the origin of the device rather than the location of processing, so that discharging an obligation with material collected from an underserved district is worth more than discharging it with metropolitan material. This converts a spatially blind market into an instrument for funding access.
- Strategy-differentiated compliance. Certificate value should be modulated by the level of value retention, so that material recycling is rated below refurbishment and qualified component harvesting. Without this, no market signal will correct the availability distortion at the top of the R-hierarchy.
- A licensing category for component harvesting and parts qualification. The intermediate strategies that retain the most value for semiconductor-dense devices currently have no regulatory home, and the informal capacity that already performs them cannot be brought into the formal system without a category to bring it into.

- Formalisation of the aggregation layer. A registration tier adapted to the real operating conditions of small aggregators (low documentation burden, digital identity, verified handover, guaranteed settlement) would capture the accessibility advantage of the informal network rather than competing with it. Indian experience with the integration of waste pickers into municipal solid waste systems provides precedent, and the evidence there indicates that inclusion outperforms displacement (Chintan, 2014; Wilson et al., 2006).
- A public capacity register. Facility throughput, current utilisation, accepted categories and buy-back prices should be published in machine-readable form in the public domain. This is the precondition for the capacity term of any execution-aware optimiser, and it is the cheapest of the five conditions to implement.

8. Designing Circular AI for Infrastructure-Constrained Settings

8.1 A Shift in Framing

Conventional circular AI optimises over the R-hierarchy and treats infrastructure as a downstream implementation detail of the recommendation. Infrastructure-Constrained Circular AI (ICCAI) reverses that ordering. Infrastructure state becomes a first-class input to the objective function, and the system optimises over the executable feasible set rather than the technically optimal one. What such a system recommends is not the best thing that could be done to the device, but the best thing that can be done to the device from where it is, given what exists, what has capacity and what is worth doing.

8.2 Architecture

ICCAI comprises four layers.

The inference layer performs conventional device-state estimation: classification, condition rating, component functionality assessment, residual-value prediction and technical R-strategy ranking. This layer is analogous to existing practice, and existing models can be reused within it directly.

The constraint layer is the novel component. It maintains a live model of the execution environment and computes the Execution Feasibility Index for each candidate device–strategy pair. Table 4 sets out the variables it must carry, their plausible sources and the gap dimensions each addresses.

Table 4. *Constraint-layer variables, data sources and the gap dimensions they address.*

Constraint variable	Data source	Update frequency	Gap dimension addressed
Facility register with licensed strategies	CPCB EPR portal; state pollution control boards	Weekly	Availability
Facility geolocation and road-network distance	Public geospatial data; routing services	Static, with route updates	Accessibility

Real-time utilisation and queue depth	Facility-reported capacity register	Daily to hourly	Capacity
Transport tariffs, consolidation state, hazardous-routing rules	Logistics providers; transport regulation	Daily	Accessibility, Viability
Secondary market prices and buy-back quotes	Marketplaces; open network protocol catalogues	Hourly	Viability
Aggregator network state and coverage	Tiered registration and handover records	Daily	Accessibility, Legibility
Compliance and documentation cost by pathway	Regulatory schedules; portal fee structures	Quarterly	Viability, Legibility

The decision layer ranks strategies by expected executed value, that is, technical retained value multiplied by the Execution Feasibility Index, rather than by technical value alone. It emits a pathway, a designated facility, an aggregation route, a dispatch window and an explicit feasibility score. Where no strategy clears the feasibility threshold, it returns a deferral with a stated reason: hold pending consolidation, hold pending capacity, or route to the highest-integrity channel available with the binding constraint recorded. A deferral is not a recommendation, and the distinction matters for reporting.

The feedback layer closes the loop that present systems leave open. Every recommendation is linked to a verified outcome, and every deviation is recorded as an execution failure attributed to a specific gap dimension. This telemetry is the raw material of the aggregate Circularity Execution Ratio, and it is what converts a recommendation system into a diagnostic instrument for infrastructure planning.

8.3 Five Design Principles

- Principle 1: Constrain, then rank. Compute feasibility first and rank desirability within the feasible set. A rank-first, filter-second architecture will systematically over-promise, because the filter stage is invariably weaker than the ranking stage and is frequently absent from deployment altogether.
- Principle 2: Geographical realism. Distance, route quality, hazardous-materials routing restrictions and consolidation state must enter the objective function directly. A system that assumes a spatially uniform facility network will generate recommendations that are correct in the metropolitan core and systematically wrong outside it. That, we argue, describes most circular AI currently deployed in India.
- Principle 3: Temporal realism about value and capacity. A device whose residual value halves in three weeks cannot be assigned to a pathway that becomes available in six. Capacity should be modelled as a stochastic resource indexed by time, not as a static property of a facility.
- Principle 4: Mixed-channel routing. The optimiser's action space must admit informal and semi-formal actors as first-class nodes with their own feasibility profiles, rather than excluding them

by definitional fiat. A model restricted to routes between registered entities is blind to the actors who perform most of the collection in the country, and its recommendations are therefore void by construction across most of the country's geography.

- Principle 5: Reflexive execution telemetry. The system should measure and expose its own execution gap. This is the principle of greatest institutional value and least technical difficulty: a national Circularity Execution Ratio disaggregated by district, device class and gap dimension would convert a diffuse infrastructure debate into a targeted investment problem, and would make phantom circularity visible in the one place it currently is not: the reporting layer.

8.4 Distributional and Governance Considerations

This design raises three governance problems. Feasibility-aware optimisation is conservatively biased against underserved regions: the hinterland has low ϕ , so an expected-value optimiser will rationally deprioritise it, potentially entrenching the very inequity the framework diagnoses. This must be countered explicitly, either by weighting the objective for equity or by separating the operational optimiser from a planning optimiser whose objective is to maximise the gain in feasibility per rupee of infrastructure investment. Second, routing to informal actors raises legitimate concerns about environmental and labour standards; the design response is graduated integration with verified handover and progressive substitution of processing functions, not unconditional inclusion. Third, capacity registers and handover records contain commercially sensitive and personally identifiable data, and the consent architecture governing them should conform to the data-minimisation and purpose-limitation norms already established in Indian digital public infrastructure practice.

9. Implications, Limitations and Future Research

9.1 Theoretical Implications

This chapter adds an explicit construct of executability to circular economy theory, and in doing so argues that the R-hierarchy is an incomplete decision framework, because it orders strategies without reference to the systems that execute them. It also offers a counterpoint to the AI-for-sustainability literature: model accuracy and system outcome are not monotonically related, and the relationship can flatten entirely in infrastructure-constrained settings. The AI-Physical Execution Gap generalises beyond e-waste to any domain in which algorithmic capacity has outstripped physical capacity: precision agriculture without cold chains, diagnostic AI without treatment capacity, credit scoring without enforcement infrastructure. We propose, as a hypothesis for testing, that capable inference paired with incapable execution is the characteristic form of the AI transition in emerging economies.

9.2 Implications for Practice and Policy

For producers and platform operators, circular targets defined over recommendation volume or certificate acquisition are not evidence of circular performance; the quantity that should be governed is a Circularity Execution Ratio computed against verified outcomes. For technology developers, the highest-value application of AI in India's circular economy is not per-device pathway recommendation but infrastructure planning: siting aggregation nodes, scheduling consolidation and identifying where marginal capacity converts the greatest volume of currently inexecutable recommendations into executable ones. For regulators, the two highest-leverage and lowest-cost measures identified here are spatial and strategy differentiation of EPR certificate value and a public machine-readable capacity register. Both address structural distortions that no degree of algorithmic sophistication can compensate for.

9.3 Limitations

Three limitations qualify these claims. First, this is a conceptual and formal framework that has not been empirically validated: the Execution Feasibility Index and the Circularity Execution Ratio are proposed instruments, and their parameters, particularly the accessibility decay parameters and the consolidation factor, require calibration against operator-level logistics and facility data that are not currently public. Second, the break-even computation in Section 6 is illustrative. Its parameter values are plausible order-of-magnitude figures rather than measured data, and it is intended to establish structure and relative magnitude rather than to estimate absolute quantities; the ranking of binding constraints it produces should be treated as a hypothesis, not a result. Third, the analysis is specific to smartphones. Its conclusions should extrapolate with adjustment to other small, semiconductor-dense devices, but large appliances, with different mass-to-value ratios and different transport economics, will exhibit materially different break-even structures and possibly a different rank-ordering of binding constraints.

9.4 Future Research

Five lines of work follow. First, empirical estimation of the Execution Feasibility Index using facility registry data, road-network distances and operator cost data, to produce the first district-level map of executable circularity in India. Second, calibration of the accessibility decay parameters by device class against observed reverse-logistics behaviour. Third, controlled comparison of conventional and constraint-aware recommendation systems on the same device population, measuring divergence in realised rather than recommended outcomes. Fourth, treatment of aggregation-node interventions as natural experiments, in order to test the consolidation predictions of Section 6.2 directly. Fifth, comparative application of the framework to other emerging markets (Nigeria, Indonesia, Brazil) to

establish whether the structure of the gap is India-specific or a general property of infrastructure-constrained circular transitions.

9.5 Consequences for Circular Business Models

The execution gap reshapes which circular business models can operate viably in an emerging market. Three consequences follow for firms. Product-as-a-service, leasing and take-back models depend on reverse logistics whose useful radius is set by the consolidation factor rather than by customer demand, so a subscription model that is profitable in a metropolitan cluster may be structurally unprofitable two hundred kilometres away in the same country. Because certificate-based compliance is spatially fungible and indifferent between refurbishment and shredding, producers face no market incentive to develop the higher-value models the R-hierarchy prefers: a firm that invests in refurbishment capacity earns no compliance premium over one that simply buys recycling certificates. And the highest-return position for a circular venture in India may not be processing at all but aggregation: a low-capital, digitally legible intermediary that converts inexecutable recommendations into executable volume and monetises the spread between informal cash offers and formal buy-back prices.

9.6 Relevance to the Sustainable Development Goals

The framework bears most directly on SDG 12, in that it separates reported circular intent from verified circular execution and supplies a measure, the Circularity Execution Ratio, that resists phantom circularity in national reporting. It bears on SDG 9 in that its principal concrete output is a district-level map identifying where marginal infrastructure investment would convert the greatest volume of currently inexecutable circularity into executable circularity. And it bears on SDG 8 in that it treats the informal sector as an execution asset to be graduated into the formal system rather than displaced from it.

10. Conclusion

The chapter opened with a village in Odisha, a working smartphone and a recommendation system that knew exactly what should be done with it. Knowing what should be done is now the cheap and abundant part of circularity in India. Being able to do it is the scarce part, and the field has repeatedly mistaken the first for the second.

The response we propose is neither to slow the deployment of AI nor to sequence infrastructure ahead of intelligence. It is to change what the intelligence optimises for. Infrastructure-Constrained Circular AI admits facility availability, distance, transport economics and real-time capacity into the objective function rather than treating them as friction to be handled later, and it closes the feedback loop that currently allows unexecuted recommendations to be counted as circular outcomes. A system built this

way recommends less and delivers more, defers honestly when deferral is the correct answer and, most valuable of all, converts every execution failure into a piece of infrastructure intelligence that is located, attributed and therefore actionable.

India's circular transition will not be decided in the recommendation layer. It will be decided on the last mile, in the aggregation shed, in the consolidated shipment, in the qualified spare part, and in the aggregator who can finally be paid and counted. What artificial intelligence is most useful for is telling us, district by district, exactly where that last mile is missing. Algorithms are extremely well suited to answering that question. It is not, at present, the question we are asking them.

Declarations

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Data availability. No new data were generated. The illustrative computation reported in Section 6.3 uses the parameter values stated in Appendix B and can be reproduced from equations (1) and (3).

Declaration on the use of generative AI. The authors used generative AI tools to assist with language editing and structural revision of the manuscript. All conceptual content, formal derivations, parameter choices, interpretations and conclusions are the authors' own, and the authors take full responsibility for the content of the published work.

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Appendix A. Derivation of the Break-Even Distance and the Minimum Viable Consignment

Equation (3) gives the net circular surplus of routing one device from an origin to a capable facility as

$$S = V_{rec} - C_{acq} - C_{agg} - C_{proc} - C_{comp} - C_{cons}/n - c_t \cdot \delta \cdot m$$

Collect the terms that are invariant with respect to distance and consolidation into a single margin term,

$$M \equiv V_{rec} - C_{acq} - C_{agg} - C_{proc} - C_{comp}$$

so that

$$S = M - C_{cons}/n - c_t \cdot \delta \cdot m \quad (A.1)$$

The break-even distance δ_{be} is the distance at which $S = 0$ for a given consolidation factor. Setting (A.1) to zero and solving for δ :

$$\begin{aligned} 0 &= M - C_{cons}/n - c_t \cdot \delta_{be} \cdot m \Rightarrow c_t \cdot \delta_{be} \cdot m = M - C_{cons}/n \\ \delta_{be} &= (M - C_{cons}/n) / (c_t \cdot m) \quad (A.2), \text{ reproducing equation (4)} \end{aligned}$$

Expression (A.2) is defined and positive only where $M > C_{cons}/n$. Taking the limit as consolidation increases without bound gives the upper envelope of the viable radius,

$$\lim_{n \rightarrow \infty} \delta_{be} = M / (c_t \cdot m) \quad (A.3)$$

which establishes the first structural property of Section 6.2: the break-even distance is increasing in n but bounded above by a quantity that depends only on the margin, the haulage tariff and the device mass. Because m is small for a smartphone, this bound is large, and the haulage term is therefore not the operative constraint over any plausible domestic distance.

The minimum viable consignment size at a fixed distance follows by setting (A.1) to zero and solving for n instead:

$$C_{cons}/n_{min} = M - c_t \cdot \delta \cdot m \Rightarrow n_{min} = C_{cons} / (M - c_t \cdot \delta \cdot m) \quad (A.4), \text{ reproducing equation (5)}$$

Expression (A.4) admits a finite positive solution only where $M > c_t \cdot \delta \cdot m$. Where $M \leq 0$, no consolidation factor renders the pathway viable, which establishes the third structural property: aggregation repairs viability failures arising from the allocation of fixed cost, but not viability failures intrinsic to the margin of the pathway itself.

Finally, differentiating (A.2) with respect to C_{comp} gives $\partial\delta_{be}/\partial C_{comp} = -1/(c_t \cdot m) < 0$, and differentiating with respect to the haulage tariff gives $\partial\delta_{be}/\partial c_t < 0$ likewise. Compliance cost and transport cost therefore enter the viable radius with the same sign, which establishes the fourth structural property.

Appendix B. Parameter Values Used in the Illustrative Computation of Section 6.3

The values below are plausible order-of-magnitude figures assembled from published road freight tariffs, observed Indian secondary-market handset prices and reported informal buy-back offers. They are stated assumptions adopted to demonstrate the structure of the framework. They are not measured operator data, and no empirical claim is made for them. All monetary values are in Indian rupees per device unless otherwise noted.

Table B1. Strategy parameters.

Strategy	V _{rec}	C _{acq}	C _{agg}	C _{proc}	C _{comp}	M = margin before transport
Refurbishment	3000	1200	150	700	100	850
Component harvesting	1400	700	150	250	100	200
Material recycling	120	60	30	40	25	-35

Table B2. Territorial regime parameters. Common to all regimes: haulage tariff $c_t = ₹6$ per tonne-kilometre; device mass $m = 0.19$ kg, giving $c_t \cdot m = ₹0.00114$ per device-kilometre.

Territorial regime	Effective road distance δ (km)	Consignment fixed cost C_{cons} (₹)	Consolidation factor n	Fixed cost per device C_{cons}/n (₹)	Haulage cost per device $c_t \cdot \delta \cdot m$ (₹)
Metropolitan core	25	400	200	2.00	0.03
Mid-periphery	180	900	40	22.50	0.21
Hinterland, no node	520	250	1	250.00	0.59
Hinterland with node	520	1400	60	23.33	0.59

Execution Feasibility Index components in Table 3 are assigned as follows. $A(r) = 1$ for refurbishment and material recycling, for which authorisation categories exist, and $A(r) = 0$ for component harvesting, for which none does. $Acc(d, r)$ is evaluated from the logistic form of Section 3.3 with the consolidation-adjusted effective distance. $Cap(r, t)$ is set at 0.80 in the metropolitan core and mid-periphery and 0.75 where dispatch is batched. $V(d, r) = 1$ where $S > 0$ and 0 otherwise, consistent with the minimum-across-chain definition. $L(d, r)$ reflects the share of the pathway passing through registered actors. Because A , V and L are effectively binary at these parameter values, ϕ is driven almost

entirely by whether an authorised executor exists and whether the pathway is value-positive, rather than by the continuous terms.